MIS772 – Predictive Analytics

T3 2025

Assignment 2 – Individual
Student name: Amruta Shailesh Kangutkar
Student number: s219349471

Executive summary (1 page)

Executive problem statement:

AirBnB is a large provider of tourist accommodations in the domestic tourism industry and wants to predict customer feedback regarding their accommodation experience at the rental properties in Sydney, Australia. A sample dataset of around 37000 rental property listings along with approximately 549000 customer reviews associated with these properties have been given by AirBnB that contain partially cleaned numerical, nominal and text variables. Data has been further cleaned with missing values removed as large number of missing values are present for a few attributes. Text processing has been done to sentiment words given in the feedback comments so that they can be processed for using in models.

The problem is that AirBnB is unsure if customer reviews are actually aligning with the what the host has described about their property listings to see if there is a significant correlation between the two. Furthermore, estimation of the review score ratings using predictors that affect them also needs to be checked and lastly, clusters need to be identified for different properties that closely related to each other so that decision making can be targeted.

Executive solution statement:

When host's property description and customer reviews were compared, a weak positive correlation was seen of 0.41 indicating that host's property description and customer ratings do not match as one would expect as seen in figure 10 below. It is recommended that hosts take this feedback and work on improving the negative ratings or fill the gaps in their description and actual reviews.

It is established from analysis that multiple regression model can be used to estimate the customer review ratings and it is found that the model's performance is weak predicting only 4.7% predictions about the customer ratings correctly even after model improvement. It is estimated that with any change in hostel (property type), review score will decrease by 2.751 units, if shared room type is changed by one unit then review score will decrease by 7.188 units and decrease by 0.035 for every unit change in host total listings count. Whereas it will increase by 1.614 if instant bookable type f is changed and by 0.002 if any change in price as seen in Fig 18. Shared accommodations including hostel type and high count of host listings seem to affect the model more so AirBnB should focus on improving these services particularly.

Lastly, property segments with cheaper, smaller accommodations with less host listings have been found to be in one segment, properties with bigger, expensive and high security deposit are found in the other segments and third segment has properties with low sentiment reviews, are pricier and hosts have higher count of listings. AirBnB should focus on these three segments to target their marketing strategies and support hosts with filling the gaps.

The benefit of the findings could help AirBnB with improving the feedback and satisfaction of customers, help hosts in improving services and in turn improve reliability and customer usage of the app.

Data exploration, pattern discovery, and preparation (2 pages)

Exploratory Data Analysis (EDA) and Data Cleaning:

Fig. 1 and 2: EDA without data cleaning- Missing values affect the visualisations in these bar charts and make them misleading, hence missing values have been removed as <5% missing values present in attributes that are going to be added in the predictive model.

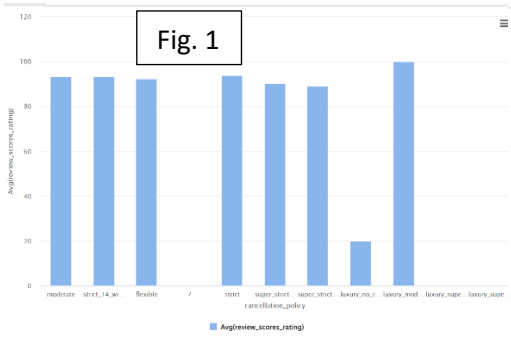


Fig. 1

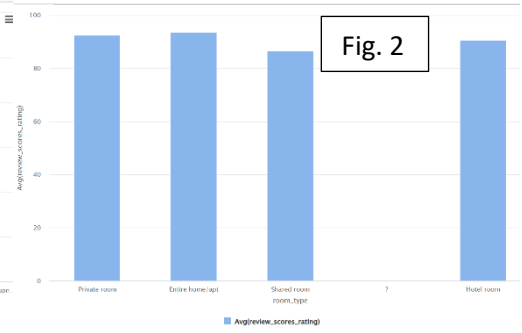


Fig. 2

Following number of missing values present in each attribute, also shows data type:

Name	Type	Missing
host_total_listings_count	Integer	155
neighbourhood	Polynominal	5246
latitude	Real	148
longitude	Real	148
property_type	Polynominal	148
room_type	Polynominal	148
accommodates	Integer	148
bathrooms	Real	170
accommodates	Integer	148
bathrooms	Real	170
bedrooms	Integer	219
price	Integer	148
security_deposit	Integer	13464
first_review	Date	10360
last_review	Date	10360
reviews_per_month	Real	10360
reviews_per_month	Real	10360
review_scores_rating	Integer	11509
review_scores_accuracy	Integer	11555
review_scores_cleanliness	Integer	11539
review_scores_checkin	Integer	11566
review_scores_communication	Integer	11542
review_scores_location	Integer	11564
review_scores_value	Integer	11571

Fig.4

Review score rating is selected as the target label for building predictive model because AirBnB needs us to estimate the review score rating of their properties.

EDA After removing missing values:

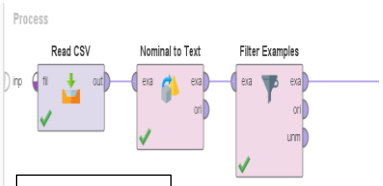


Fig.5

Fig. 5 shows nominal data type converted to text for processing documents and sentiment analysis so that the text is not treated as categorical.

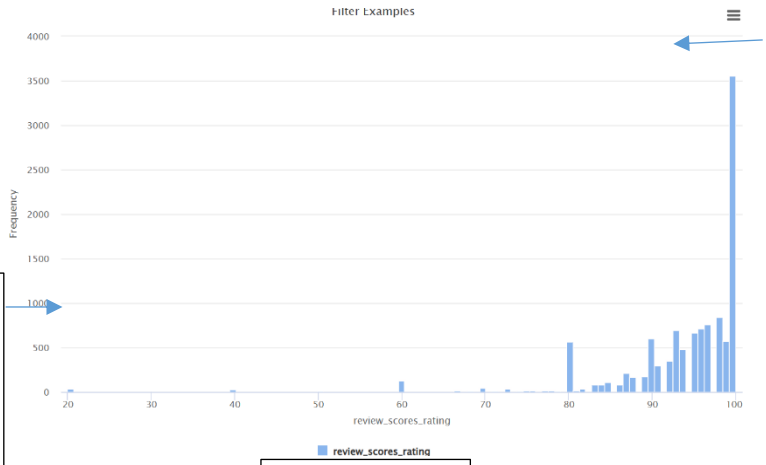


Fig. 6

Fig. 6 shows histogram of target variable shows highly right skewed data showing most ratings are on the higher side (90-100) compared to lower ratings which could either show high satisfaction or low engagement from customers who have lower ratings to give.

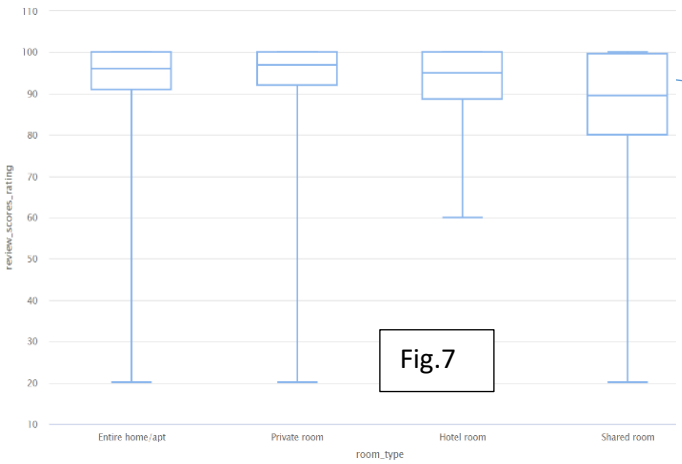


Fig.7

Figure 7 shows boxplot of review score ratings with room types and shows most ratings are at higher ranges mostly between 80-100. The ratings for shared rooms are more varied and have slightly lower average ratings than other three-room types. Thus, room type may not be a very good predictor of review score ratings which further needs to be assessed using correlation.

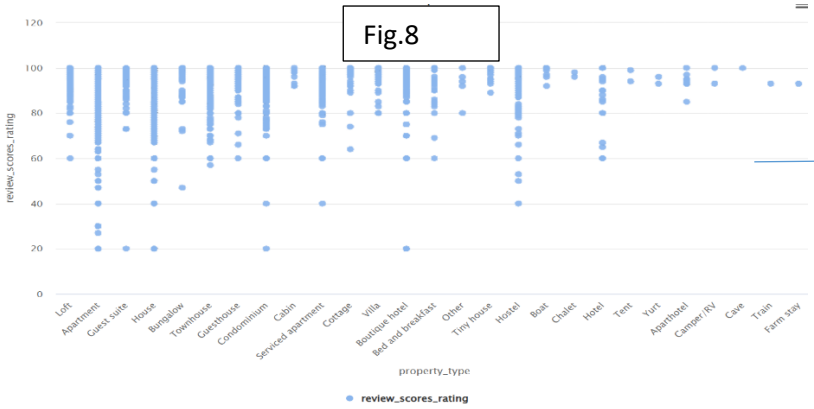
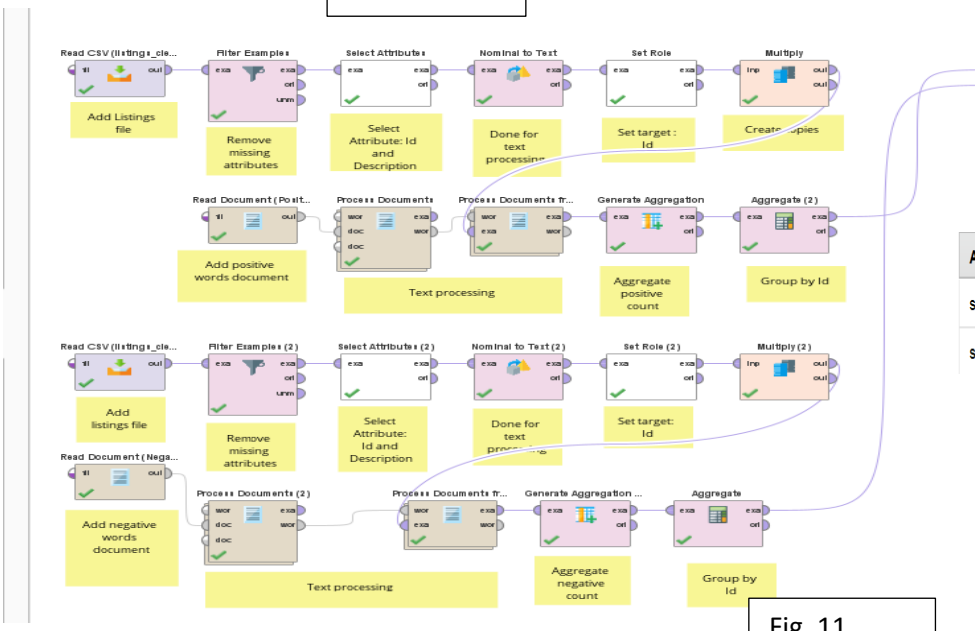


Figure 8. Scatter plot between property type and reviews score rating shows most review score ratings to be of higher values between 80-100. There are many outliers and data is very scattered for all property types. It may not be a very strong predictor for review rating score.

Task A

Fig.9



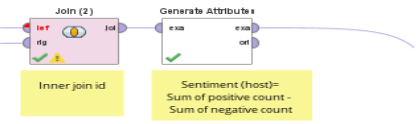


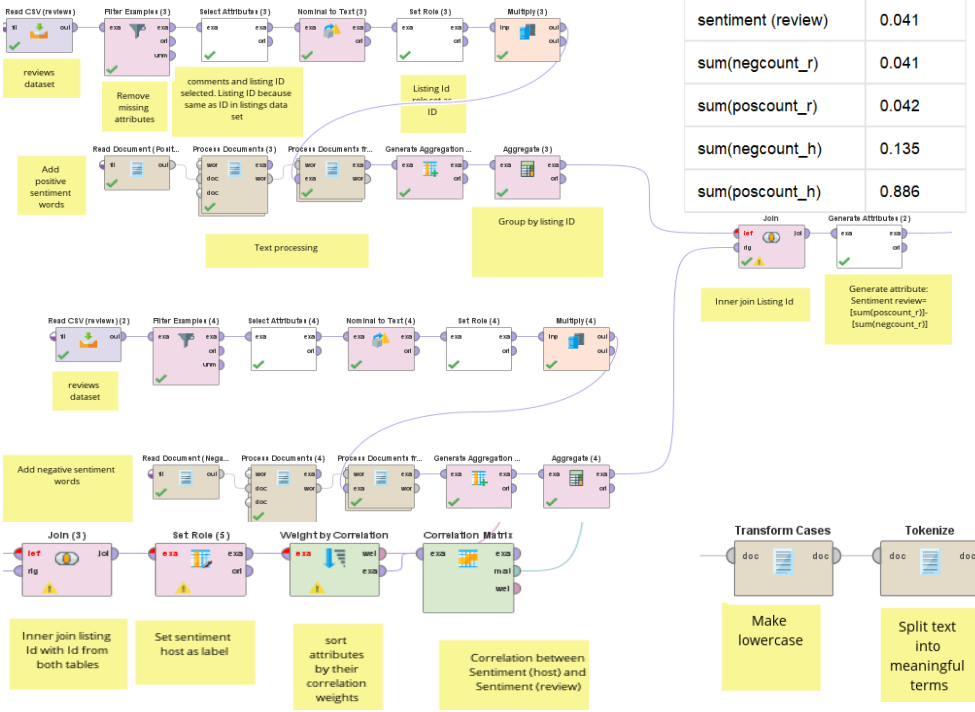
Fig. 10: Correlation matrix

Attributes	sentiment (host)	sentiment (review)
sentiment (host)	1	0.041
sentiment (review)	0.041	1

All tables and processes on this page show the working for to determine if there is a significant correlation between the raw sentiment scores of the host's property description and that of feedback/reviews from the customers by doing sentiment analysis. Listings and Reviews dataset have been combined along with positive and negative sentiment words. Text analytics (syntactic analysis) has been used to extract the positive and negative sentiment words. From the correlation matrix (Fig. 10) it is evident that there is weak positive correlation between the raw sentiment scores as the value is 0.041 indicating that the hosts' description of the properties does not match with the customer review.

Fig. 11

attribute	weight
sentiment (review)	0.041
sum(negcount_r)	0.041
sum(poscount_r)	0.042
sum(negcount_h)	0.135
sum(poscount_h)	0.886



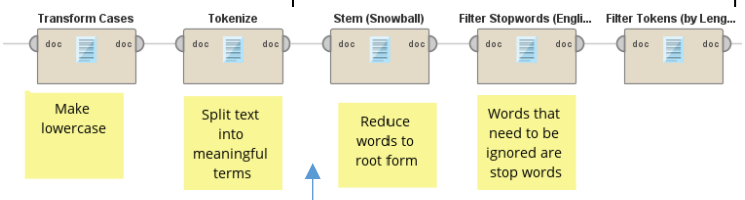


Fig. 12: Process documents (parsing) steps above

Multiple linear regression model is the model of choice for estimating the review score ratings for the following reasons:

1. The target attribute is numerical.
2. Linear regression models are used when are predicting values (estimating review score ratings in this case).
3. There are multiple independent variables here that can help in predicting the value of one dependent variable which is review score ratings.

Correlation matrix to determine which attributes will go into the model based on highest correlation values:

Fig. 13

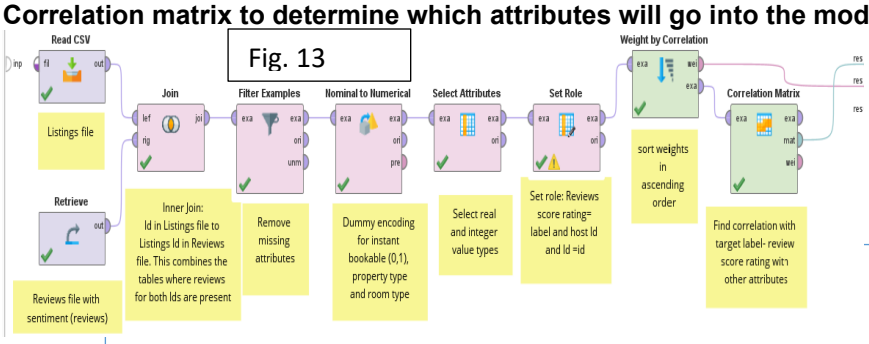


Fig. 14

Attributes	propert...	propert...	propert...	propert...	propert...	propert...	propert...	propert...	propert...	propert...	propert...	propert...
property_type = Apartment	1	-0.128	-0.689	-0.174	-0.272	-0.079	-0.190	-0.161	-0.041	-0.088	-0.088	-0.120
property_type = Loft	-0.128	1	-0.055	-0.014	-0.022	-0.006	-0.015	-0.013	-0.003	-0.007	-0.004	-0.010
property_type = House	-0.689	-0.055	1	-0.075	-0.117	-0.034	-0.082	-0.069	-0.018	-0.038	-0.024	-0.052
property_type = Guest suite	-0.174	-0.014	-0.075	1	-0.030	-0.009	-0.021	-0.017	-0.004	-0.010	-0.006	-0.013
property_type = Townhouse	-0.272	-0.022	-0.117	-0.030	1	-0.013	-0.032	-0.027	-0.007	-0.015	-0.009	-0.020
property_type = Bungalow	-0.079	-0.006	-0.034	-0.009	-0.013	1	-0.009	-0.008	-0.002	-0.004	-0.003	-0.006
property_type = Condominium	-0.190	-0.015	-0.082	-0.021	-0.032	-0.009	1	-0.019	-0.005	-0.010	-0.007	-0.014
property_type = Guesthouse	-0.161	-0.013	-0.069	-0.017	-0.027	-0.008	-0.019	1	-0.004	-0.009	-0.006	-0.012
property_type = Cabin	-0.041	-0.003	-0.018	-0.004	-0.007	-0.002	-0.005	-0.004	1	-0.002	-0.001	-0.003
property_type = Cottage	-0.088	-0.007	-0.038	-0.010	-0.015	-0.004	-0.010	-0.009	-0.002	1	-0.003	-0.007
property_type = Bed and breakfast	-0.088	-0.004	-0.024	-0.006	-0.009	-0.003	-0.007	-0.006	-0.001	-0.003	1	-0.004
property_type = Serviced apartment	-0.120	-0.010	-0.052	-0.013	-0.020	-0.006	-0.014	-0.012	-0.003	-0.007	-0.004	1
property_type = Villa	-0.080	-0.006	-0.034	-0.009	-0.014	-0.004	-0.009	-0.008	-0.002	-0.004	-0.003	-0.006
property_type = Boutique hotel	-0.123	-0.010	-0.053	-0.013	-0.021	-0.006	-0.015	-0.012	-0.003	-0.007	-0.004	-0.009
property_type = Farm stay	-0.012	-0.001	-0.005	-0.001	-0.002	-0.001	-0.001	-0.001	-0.000	-0.001	-0.000	-0.001

Fig.15

attribute	wei...
review_scores_location	0.436
host_total_listings_count	0.130
instant_bookable = f	0.116
instant_bookable = t	0.116
sentiment (review)	0.109
sum(poscount_r)	0.097
room_type = Shared room	0.094
longitude	0.076
reviews_per_month	0.068
property_type = Hostel	0.063
property_type = Hotel	0.044
latitude	0.044
bedrooms	0.039
price	0.039
property_type = Apartment	0.038
sum(negcount_r)	0.036

Feature engineering: In the figure 13, listings and review files with sentiment scores are combined, and correlation matrix (fig. 14) is created to check which predictors affect the target label the most, and accordingly which will be added to the model for estimating the review score ratings. There are attributes related to review scores already from which they have been calculated. These values are not used in model development due to being too related to the target and can cause data leakage. Thus, the attributes selected to go into the model from Fig. 15 are instant bookables f, room type (shared room), property type (hostel), host total listings count, price and sentiment (review) due to comparatively higher weights of Pearson's correlation with the target. Bedrooms and bathrooms are not included due to multicollinearity with the accommodates attribute. Accommodates attribute is not added as well due to weaker correlation. Latitude and longitude not added as they are not good predictors for the target and have limited interpretability.

Multiple Linear Regression model:

Fig.16

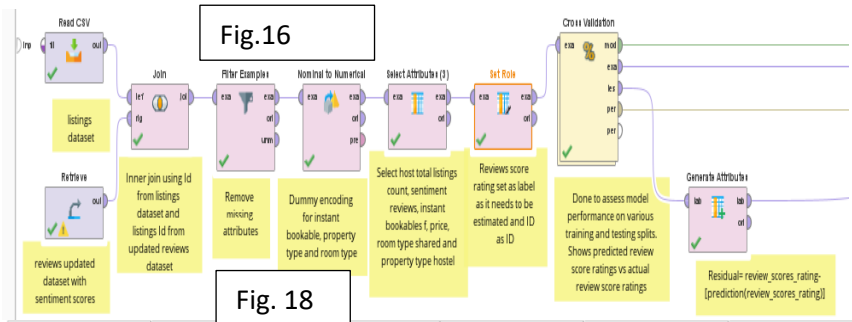


Fig.17

```
PerformanceVector:
root_mean_squared_error: 8.739 +/- 0.163 (micro average: 8.741 +/- 0.000)
squared_correlation: 0.047 +/- 0.010 (micro average: 0.046)
```

From Fig. 16 and 17, it is evident that R^2 (coefficient of determination) shows that the model can explain only 4.7% of variance from the predictors which is a very weak predictor of model performance.

Fig. 18

Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value	Code
property_type = Hostel	-2.751	1.218	-0.023	0.820	-2.258	0.024	**
room_type = Shared r...	-7.188	0.994	-0.073	0.818	-7.228	0.000	***
instant_bookable = f	1.614	0.168	0.089	0.960	9.593	0	***
host_total_listings_co...	-0.035	0.003	-0.123	0.889	-12.813	0	***
price	0.002	0.000	0.076	0.920	8.057	0.000	***
sentiment (review)	0.004	0.000	0.099	0.985	10.811	0	***
(Intercept)	92.313	0.156	?	?	592.882	0	***

Multiple regression equation: Review scores ratings= 92.31- (2.751)*(property type hostel)- (7.188)* (room type shared room) + (1.614)* (instant bookable F) – (0.035)* (host total listings count) + (0.002)* (price) + (0.004)* (sentiment review) + e

Interpretation of the equation and estimates of review score ratings: It suggests that with any changes in property type-hostel, review score ratings will decrease by 2.751, if changes in shared room type considered then decrease by 7.188. Review score ratings will also increase by 1.614 if instant bookable type f is changed, decrease by 0.035 for 1 unit change in host total listings count, and increase by 0.002 if changes in price and by 0.004 if changes in sentiment review.

Clustering models:

For determining the most number of distinct property segments in the given AirBnB data, clustering models were developed. This is because the AirBnB data needs to be grouped in a way that all data points in the same group are similar to each other more than with other groups. In this model, target label is not determined and is unsupervised learning where no dependent variable is present. Only numerical predictor attributes which are not interrelated were chosen for this.

Fig. 19

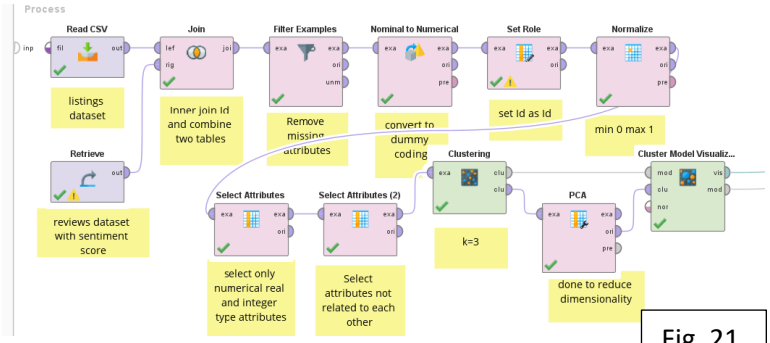


Fig. 20

Cluster Model

```
Cluster 0: 8182 items
Cluster 1: 2690 items
Cluster 2: 662 items
Total number of items: 11534
```

Fig. 20 shows the number of items in each of the 3 clusters.

Fig 19 shows k-means clustering method is used for finding the centres of the groups. K=3 was chosen (see model evaluation below).

Fig. 21

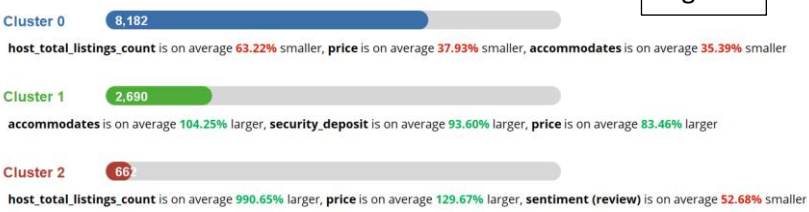
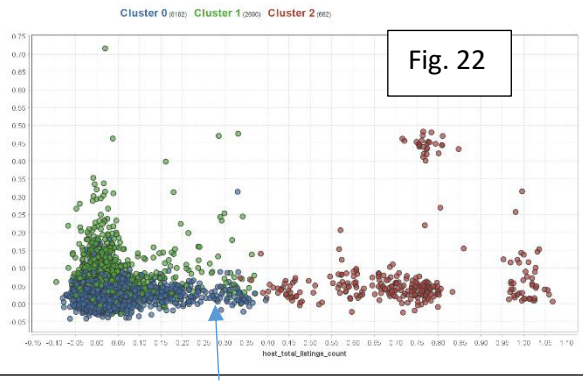


Fig. 22



Based on the three clusters in fig 21, the most meaningful segments in the AirBnB data are first segment with smaller accommodations that cost less, and hosts have less listings, second segment is bigger and costlier accommodations with high security deposit and third cluster with low sentiment review scores, costly and listings with hosts having higher counts of accommodations.

In Fig. 22, the cluster cohesion and cluster separation seen in one of these plots between host total listings count and price is weak. The three clusters are getting influenced by other attributes as well which are not a part of cluster.

Fig. 23

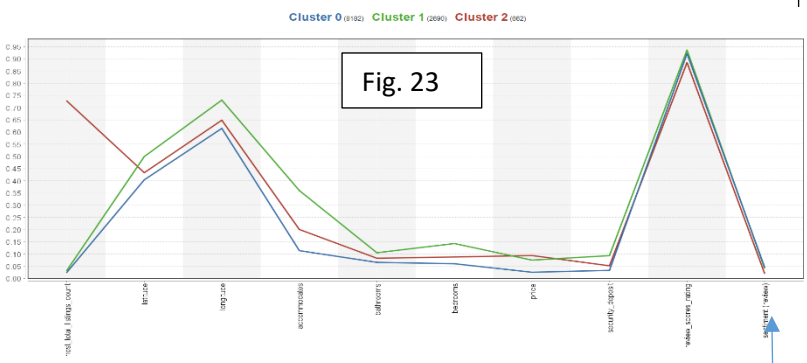
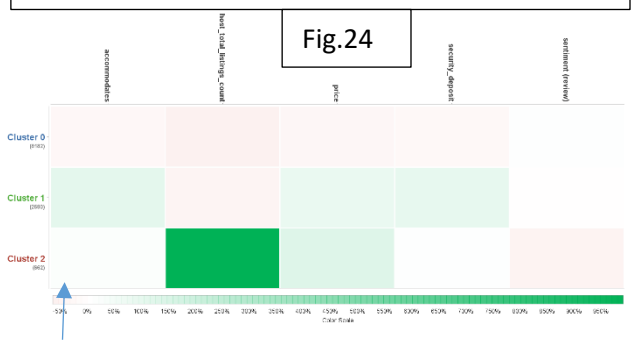


Fig.24



From cluster model visualiser in fig 23, the clusters with different segments are confirmed showing listings with low host listings count, accommodate less people with lower prices. There is an overlap between all clusters for security deposit, review score rating and sentiment reviews with sentiment scores for cluster 1 being slightly higher and cluster 2 being slightly lower than cluster 0. This is confirmed in Fig 24 heatmap.

Model evaluation and improvement (2 pages)

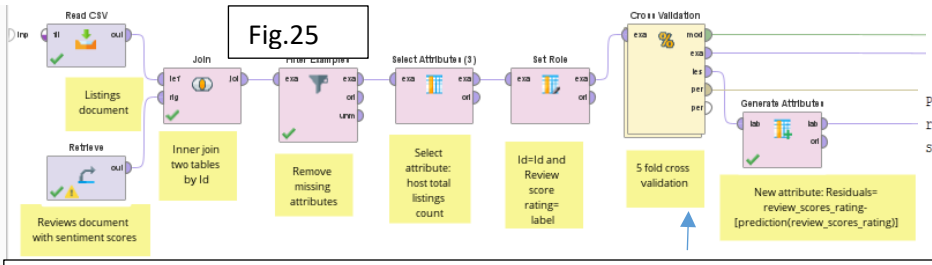


Fig.26

```
PerformanceVector:
root_mean_squared_error: 8.873 +/- 0.174 (micro average: 8.874 +/- 0.000)
squared_correlation: 0.017 +/- 0.005 (micro average: 0.016)
```

From fig 25 and 26, first simple linear regression model was built and evaluated. It was not chosen as the model of choice due to poorer performance (1.7% values correctly predicted) as shown in the figures above.

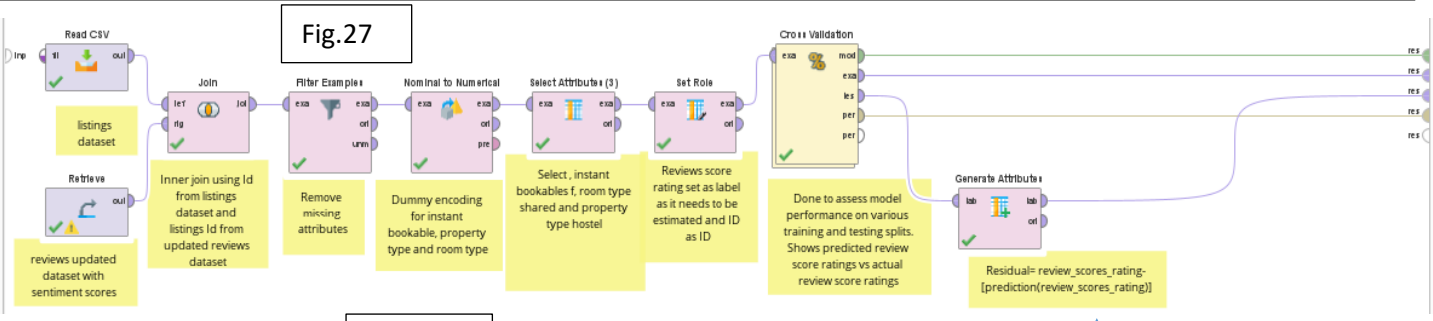


Fig.28

```
PerformanceVector:
root_mean_squared_error: 8.855 +/- 0.168 (micro average: 8.856 +/- 0.000)
squared_correlation: 0.022 +/- 0.009 (micro average: 0.020)
```

(Fig 27, 28) To improve performance of the predictive model, more attributes are added to the model. Feature engineering (weighing) is done by building correlation matrix and attributes with highest correlation with review score ratings are selected. It is ensured, attributes that can cause leakage or are multicollinear are not added. When features such as host total listings count, sentiment reviews, instant bookables f, price, room type shared and property type hostel were selected, the model performed slightly better than when only the attributes like instant bookables f, room type shared and property type hostel were selected. The model performance with only these three attributes was only 2.2% (in pictures above) compared to the earlier mentioned model that predicted 4.7% values correctly.

Evaluating the multiple linear regression model built above, it is seen that the tolerance (1-R^2 of all the selected attributes is high which indicates that there is no high correlation between the selected attributes. Thus, it is recommended to keep all these attributes in the model. Furthermore, since all attributes have p- value less than 0.05, they are all statistically significant and can be kept in the model.

Fig 29

Row N...	id	review_scores_rating	prediction(review_scores_rating)	property...	room_type ...	instant_boo...	host_total_li...	price	sentiment (r...	Residual
1	66158	78	94	0	0	1	2	80	37	-16
2	86853	91	95	0	0	0	0	199	528	-4
3	87634	96	95	0	0	1	1	89	343	1
4	118658	87	94	0	0	1	1	225	12	-7
5	210439	96	95	0	0	1	1	122	94	1
6	226202	90	94	0	0	1	1	129	14	-4
7	255090	82	93	0	0	0	1	104	78	-11
8	259857	97	96	0	0	0	1	110	760	1
9	271182	100	95	0	0	1	1	143	168	5
10	288671	96	95	0	0	1	1	220	137	1
11	299749	94	93	0	0	0	2	150	22	1
12	366856	89	94	0	0	0	3	134	422	-5
13	366857	91	94	0	0	0	3	140	429	-3
14	379498	93	95	0	0	1	2	220	244	-2

The adjacent figure 29 shows the actual review scores rating values vs the predicted review score rating values after using the multiple regression model that predicts 4.7% values correctly. Residuals show the difference between the scores. There are many negative and positive values indicating model is both over and underestimating values.

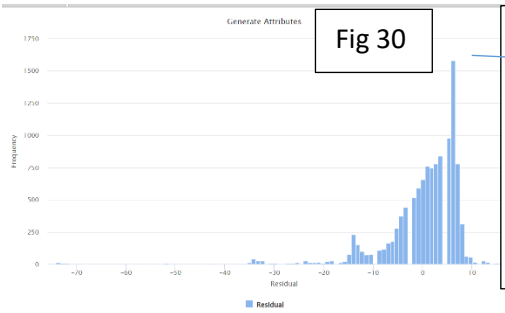
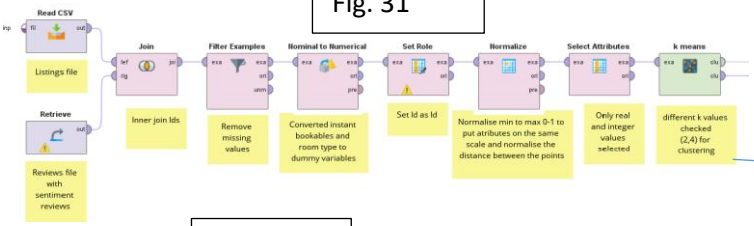


Fig 30

Histogram (Fig 30) of residuals shows left skewness towards negative values indicating model may be overestimating values. A histogram of residuals closer to zero are better outcomes than the ones away from zero. The histogram is approximately bell shaped so we can stick to linear regression model that has been used.

Clustering models:

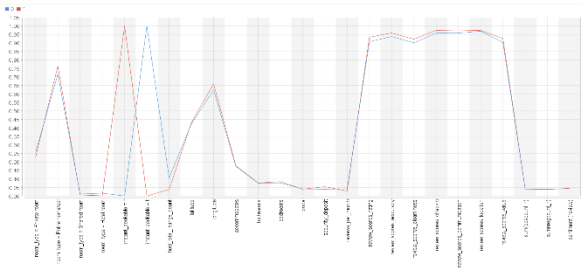
Fig. 31



Clustering models were built with various k values (2 and 4) and model was visualised as follows:

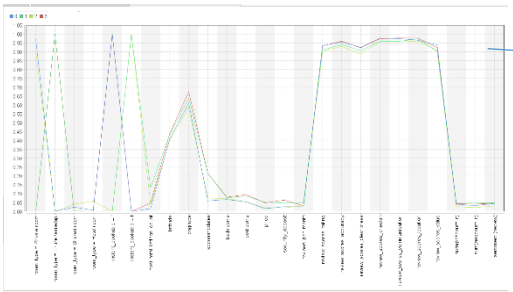
K=2

Fig. 32



k=4

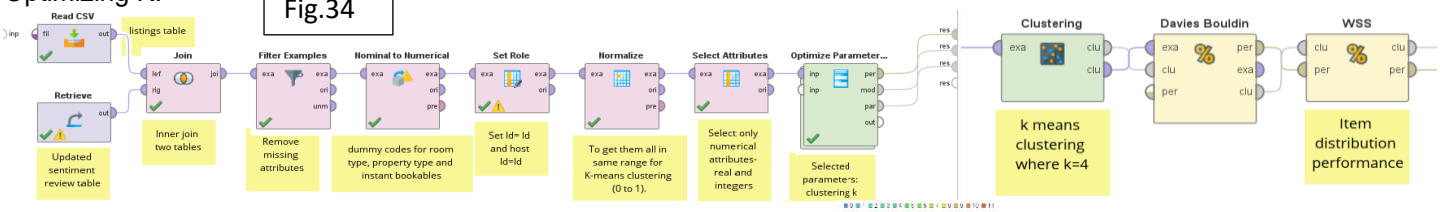
Fig.33



Clusters are bit more separated than at k value =2 indicating higher k would be better for clustering. It was further checked by Davies Bouldin score and elbow method.

Optimizing K:

Fig.34



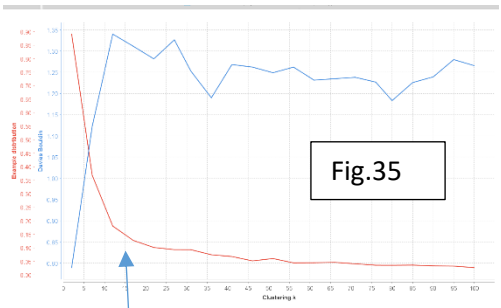


Fig.35

ParameterSet

```
Parameter set:

Performance:
PerformanceVector [
*****Davies Bouldin: 1.340
-----Example distribution: 0.181
]
Clustering.k    = 12
```

Fig. 36

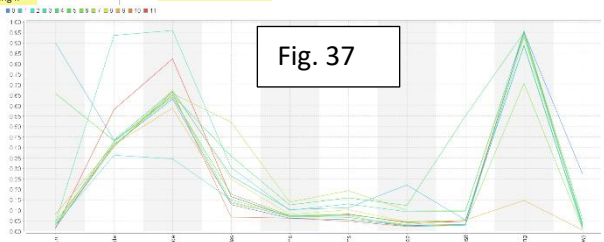


Fig. 37

Performance was plotted against parameters and Davies Bouldin score indicated k=3 is the most appropriate value of k and it is 7 using elbow method too. K=7 is not considered due to practicality of the model. It indicates that if k value is taken as 3, the cohesion and separation between the clusters will be better. Further, only relevant attributes were chosen to improve model. Davies Bouldin score is not very close to 0 (1.34) indicating weaker clusters with overlaps.

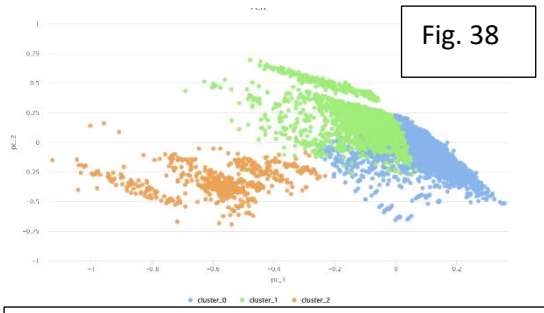


Fig. 38

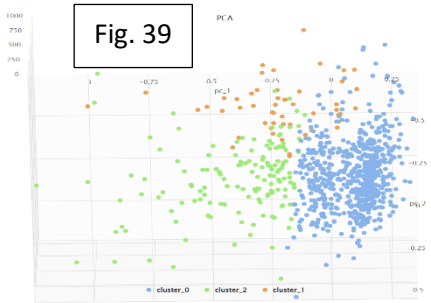


Fig. 39

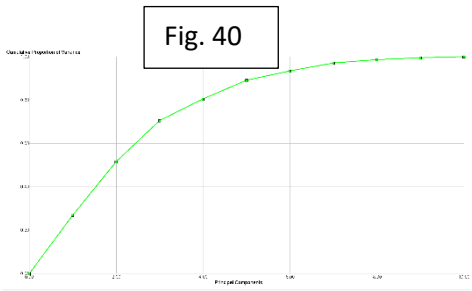


Fig. 40

Fig 38 and 39: To ensure independence of principal components and their sequencing in the order of their variance and to eliminate multicollinearity in data, Principal Component Analysis has been done. PC1 is explaining the most variance in the data followed by PC2 and the rest. Unclustered data points look higher. It is also visualised in 3D and considerable overlapping in the clusters from all angles is found.

Fig. 40: Plot shows cumulative variance for all principal components. First 4 component capture around 80% of the variance in data.